

Question	Answer	Mark	Mark scheme	Additional guidance
23	264	P1	correct substitution into the volume formula, eg $56.8 = \frac{1}{3} \times \pi \times r^2 \times 3.6$	<i>AOB</i> does not need to be the subject of the equation
		P1	completes process to find base radius or the value of r^2 , eg $r = \sqrt{\frac{56.8 \times 3}{\pi \times 3.6}}$ (=3.88158...) or $r^2 = \frac{56.8}{1.2\pi}$ (=15.066)	
		P1	Uses Pythagoras to find the sloping length, eg $\sqrt{"3.88 \dots" ^2 + 3.6^2}$ (=5.29.....)	
		P1	process to find an equation in <i>AOB</i> , eg $\pi \times "3.88" \times "5.29" = \frac{AOB}{360} \times \pi \times "5.29" ^2$ or $\frac{AOB}{360} \times \pi \times 2 \times "5.29" = 2 \times \pi \times "3.88"$ or $\frac{AOB}{360} \times "5.29" = "3.88"$	
		A1	answer in the range 263.9 to 264.1	